Data Pipeline for a Manufacture-to-Customer Online Shopping Platform

- **Concept:** Manufacture-to-Consumer (M2C) Direct Shopping.

- **Goal:** Building high-quality, processed data pipeline to help entrepreneurship and startup make data-driven decisions.

Background & Problem Statement

Manufacture-to-customer (M2C) startups aim to sell products directly from manufacturers to customers, bypassing middlemen. While this model reduces costs, it introduces challenges:

- Managing inventory and stock levels in real-Time.
- Tracking orders and shipments efficiently.
- Understanding customer behavior and preferences.
- Generating insights to support business decisions.

Currently, many small startups rely on spreadsheets or manual tracking, leading to errors, delays, and lost sales.

Project Objective

The goal of this project is to **design and implement a data pipeline** that:

1. **Ingests data** from multiple sources: customers, orders, products, and manufacturers.
2. **Cleans, transforms, and enriches data** for analytics.
3. **Stores data in a centralized database** for easy access.
4. **Generates actionable insights** to help startups make data-driven decisions.
5. **Simulates recommendations and alerts** for inventory and customer engagement.

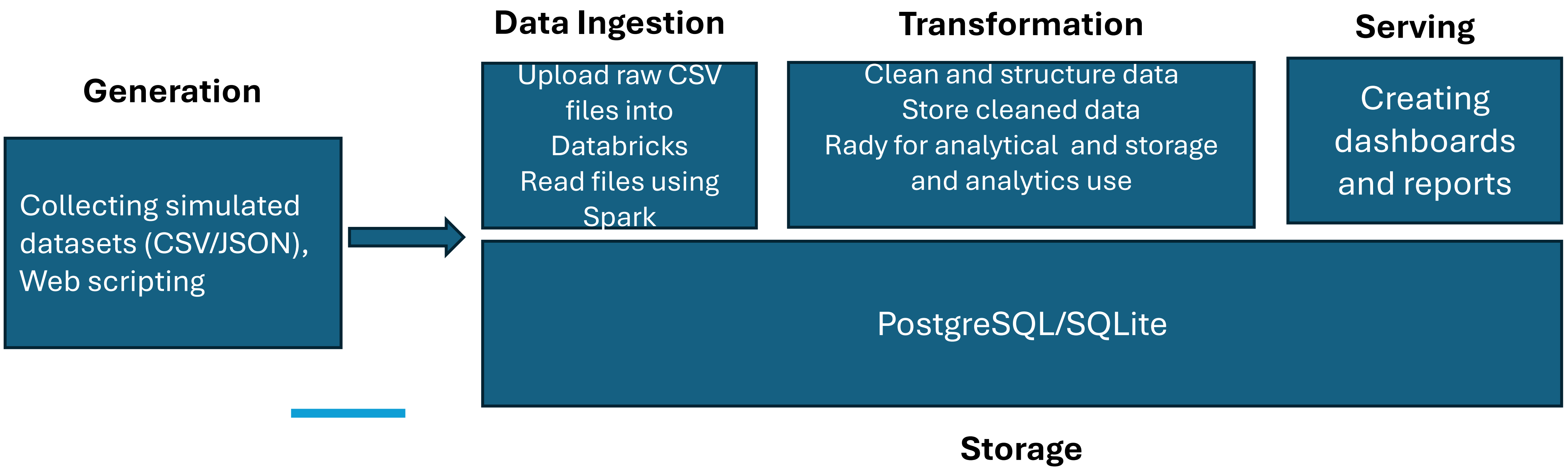
Alignment with Data Engineering life cycle

Phase	Implementation
Generation	Collecting simulated datasets (CSV/JSON) for customers, products, orders, and manufacturers.
Data Ingestion	Loading data into staging using Python scripts and batch processing.
Data Processing & Transformation	Cleaning, merging, and calculating metrics; preparing data for storage and analytics.
Data Storage	Designing relational database schema in PostgreSQL/SQLite.
Data Analysis & Visualization	Creating dashboards and reports

Possible Data Sources

Data Item	Importance	Accessible
Customers Data	To analyze customer behavior, region-based sales, repeat purchases	Simulate CSV/JSON datasets
Products	Central to sales, pricing, and category analysis	Simulate CSV/JSON datasets
Orders	Main transactional dataset	Simulate CSV/JSON datasets
Order_Items	To connect products with orders (many-to-many relationship)	Simulate CSV/JSON datasets
Inventory	For supply chain health.	Simulate CSV/JSON datasets
Consumer Price Index	Measure general price inflation in the economy	https://fred.stlouisfed.org/series/CPIAUCSL (CSV file)

Alignment with the Data Engineering Lifecycle



Tools & Technologies

Layer	Tool / Technology
Data Ingestion	Python (pandas, CSV/JSON)
Data Processing	Python (pandas), SQL
Data Storage	PostgreSQL / SQLite
Analytics / Visualization	Python (matplotlib), Streamlit
Collaboration	GitHub, Slack

Teamwork Policy

1- Roles & Responsibilities:

Each team member will have defined roles: Data Ingestion, Processing, Database Design, Analytics/Dashboard, Documentation/Presentation.

2- Code Collaboration:

Use GitHub for version control and collaborative development.

Code reviews for quality assurance before merging.

3- Communication & Meetings:

- Weekly team meetings to discuss progress, challenges, and next steps.
- Use Slack/Discord for day-to-day communication.

4- Conflict Resolution:

Open discussion and role reassignment if workload is unbalanced.

Document contributions.

5- Documentation:

Maintain a shared project report and README with clear explanation of pipeline, methods, and insights.

Challenges might be happened

Handle different Data sources (CSV vs JSON vs API):

Integration Complexity

Time Constraints